N.Yugesh(192424329)

Set 6:

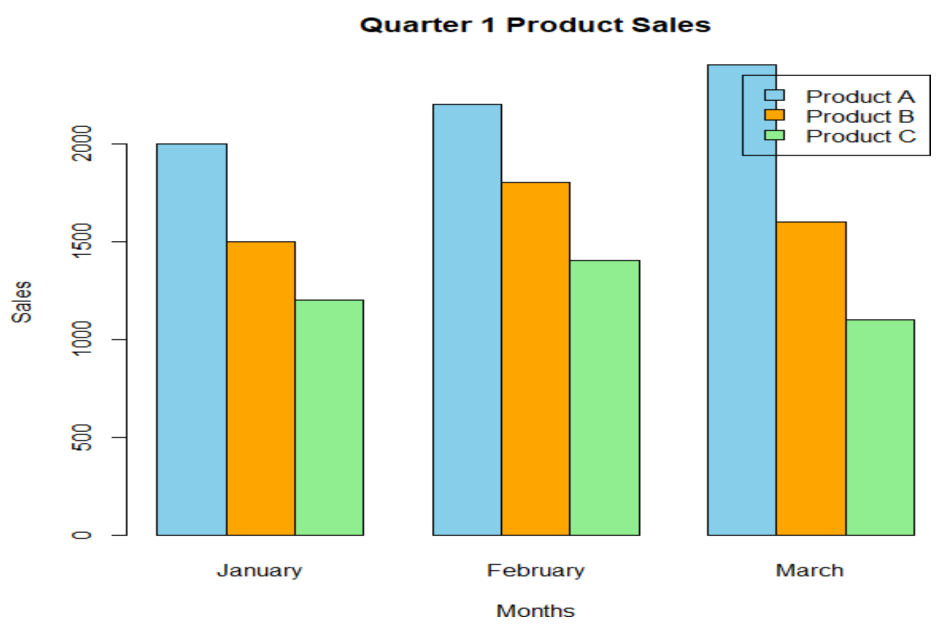
Product Sales Analysis

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Code:

```
# Product Sales Data
sales <- matrix(c(
  2000,1500,1200,
  2200,1800,1400,
  2400,1600,1100
), nrow=3)
colnames(sales) <- c("January","February","March")
rownames(sales) <- c("Product A","Product B","Product C")
barplot(
sales,
beside=TRUE,
col=c("skyblue","orange","lightgreen"),
main="Quarter 1 Product Sales",
xlab="Months",
ylab="Sales",
legend.text=rownames(sales)
)
```

Output:

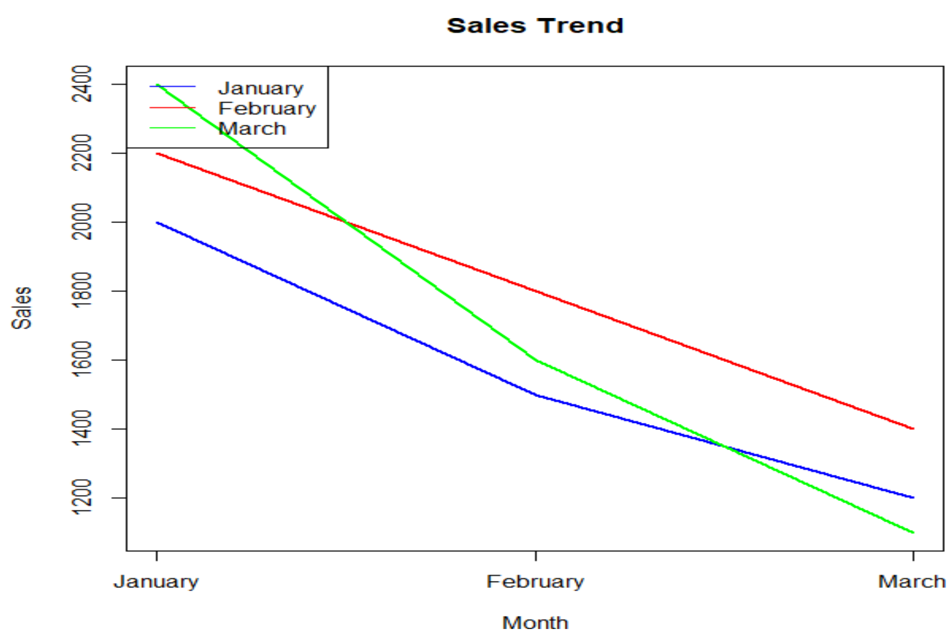


2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Code:

```
January <- c(2000,1500,1200)
February <- c(2200,1800,1400)
March <- c(2400,1600,1100)
Month <- 1:3
matplot(
Month,
cbind(January,February,March),
type="l",
lty=1,
lwd=2,
col=c("blue","red","green"),
xaxt="n",
xlab="Month",
ylab="Sales",
main="Sales Trend"
)
axis(1,
at=1:3,
labels=c("January","February","March"))
legend(
"topleft",
legend=c("January","February","March"),
col=c("blue","red","green"),
lty=1)
```

output:



3. Build a table to show the monthly sales data for each product. Label the table elements.

Code:

```r
Product <- c("Product A","Product B","Product C")

January <- c(2000,1500,1200)
February <- c(2200,1800,1400)
March <- c(2400,1600,1100)

sales_table <- data.frame(
Product,
January,
February,
March
)

print(sales_table)
```

output:

```
    Product January February March
1 Product A    2000     2200  2400
2 Product B    1500     1800  1600
3 Product C    1200     1400  1100
>
```

Set 7 Customer Demographics Analysis

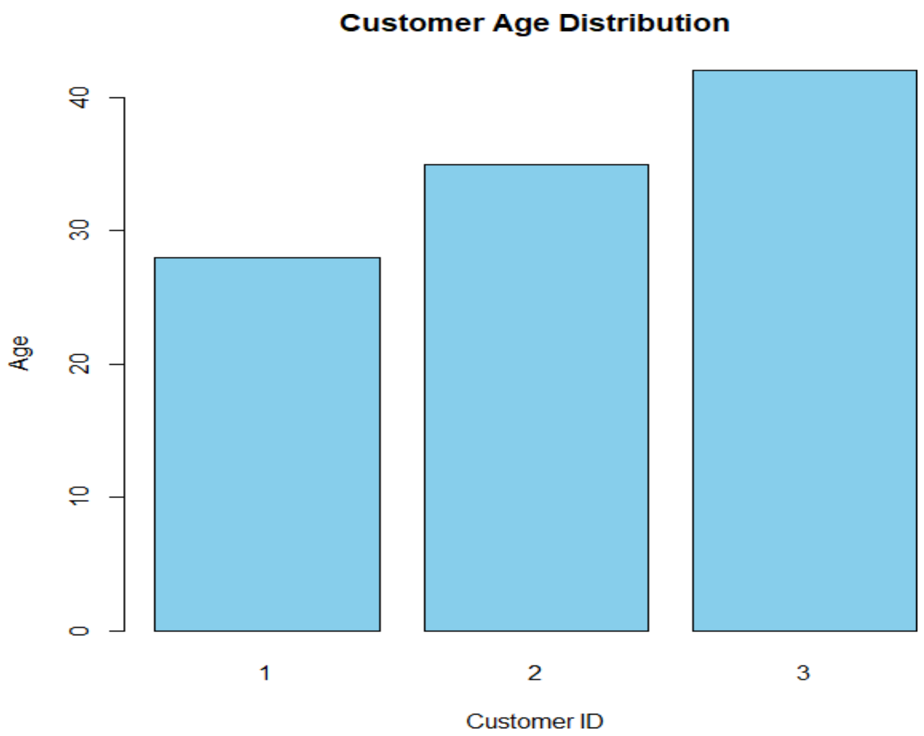
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Code:
```
# Customer Demographics Data


Customer_ID <- c(1,2,3)
Age <- c(28,35,42)

barplot(
Age,
names.arg=Customer_ID,
col="skyblue",
main="Customer Age Distribution",
xlab="Customer ID",
ylab="Age"
)
```

Output:



2. Generate a pie chart to represent the distribution of customers by gender.

Code:

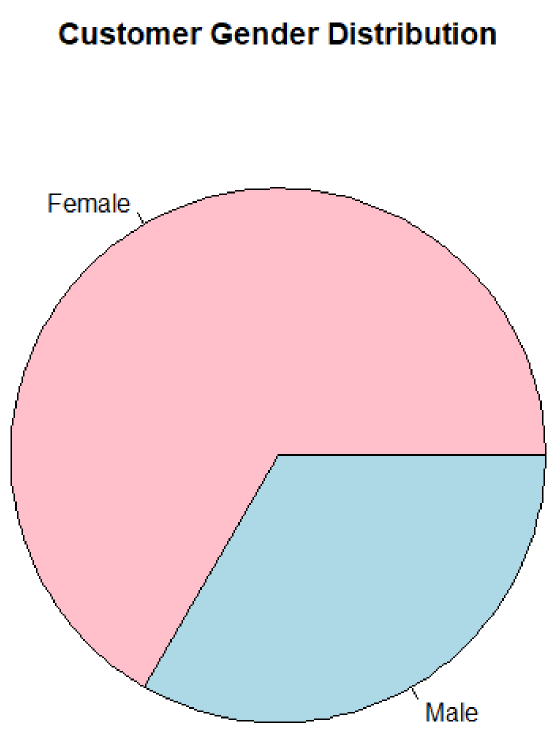
```
# Customer Gender Data

Gender <- c("Female","Male","Female")

gender_count <- table(Gender)

pie(
gender_count,
col=c("pink","lightblue"),
main="Customer Gender Distribution"
)
```

Output:



3. Build a table to show the demographic information for each customer. Label the table elements.

Code:

```
Customer_ID <- c(1,2,3)
Age <- c(28,35,42)
Gender <- c("Female","Male","Female")
Income <- c(50000,60000,75000)

customer_data <- data.frame(
Customer_ID,
Age,
Gender,
Income
)

print(customer_data)
```

output:

```
  Customer_ID Age Gender Income
1           1  28 Female  50000
2           2  35   Male  60000
3           3  42 Female  75000
>
```

Set 8 Employee Performance Analysis

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Code:

```
# Employee Performance Scores

Performance <- c(85, 92, 78, 90, 76)

hist(
  Performance,
  col = "skyblue",
  main = "Distribution of Employee Performance Scores",
  xlab = "Performance Score",
  ylab = "Frequency"
)
```

Output:

2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

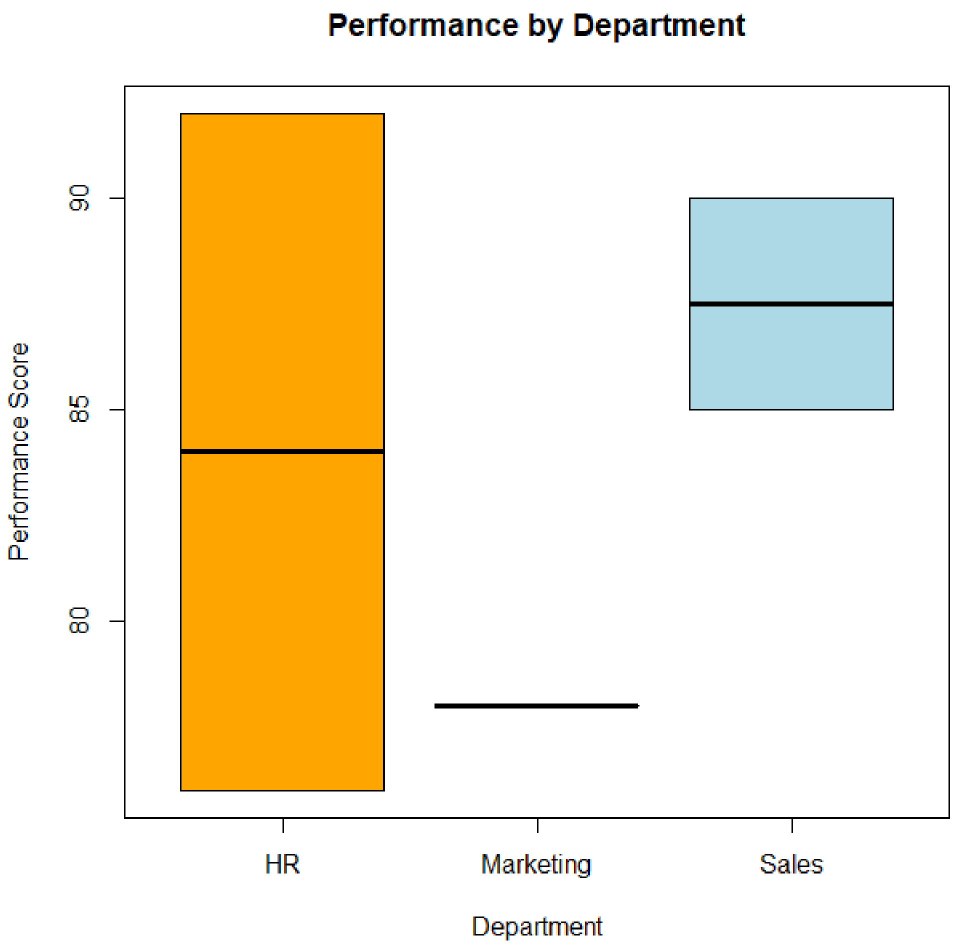
Code:

```
# Employee Data

Department <- c("Sales", "HR", "Marketing", "Sales", "HR")
Performance <- c(85, 92, 78, 90, 76)

boxplot(
  Performance ~ Department,
  col = c("orange", "lightgreen", "lightblue"),
  main = "Performance by Department",
  xlab = "Department",
  ylab = "Performance Score"
)
```

Output:



3. Build a table to display the performance data for each employee. Label the table elements.

code:

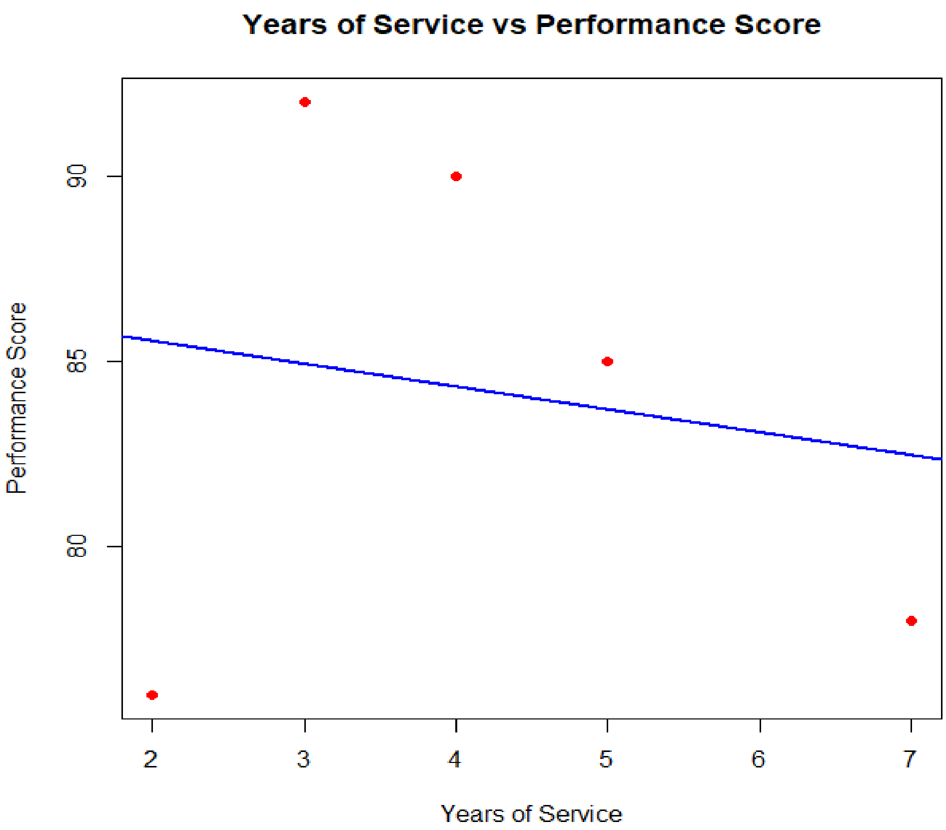
```
# Years of Service vs Performance

Years_Service <- c(5, 3, 7, 4, 2)
Performance <- c(85, 92, 78, 90, 76)

plot(
  Years_Service,
  Performance,
  pch = 19,
  col = "red",
  main = "Years of Service vs Performance Score",
  xlab = "Years of Service",
  ylab = "Performance Score"
)

abline(lm(Performance ~ Years_Service), col = "blue", lwd = 2)
```

output:



set 9 Online_Learning_Activity

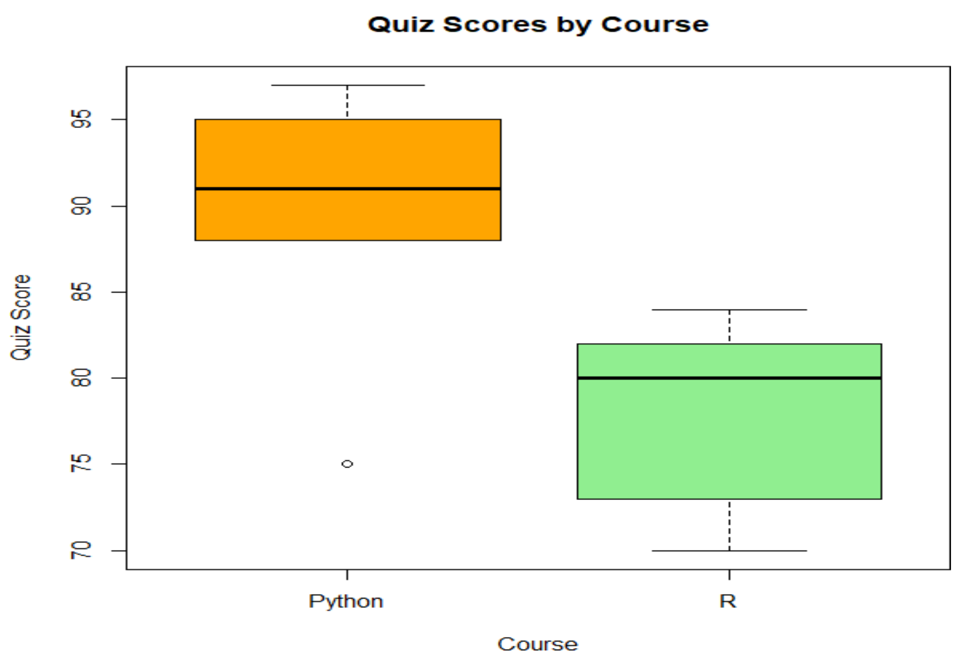
1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by

Course. Add title and labels. Interpret student performance.

Code:

```
# Online Learning Activity Dataset
Course <- c("Python","R","Python","R","Python","R","Python","R","Python","R")
Quiz_Score <- c(75,82,88,70,95,80,91,73,97,84)
# Histogram
hist(
Quiz_Score,
col="skyblue",
main="Histogram of Quiz Scores",
xlab="Quiz Score",
ylab="Frequency"
)
# Box Plot
boxplot(
Quiz_Score ~ Course,
col=c("orange","lightgreen"),
main="Quiz Scores by Course",
xlab="Course",
ylab="Quiz Score"
)
```

Output:

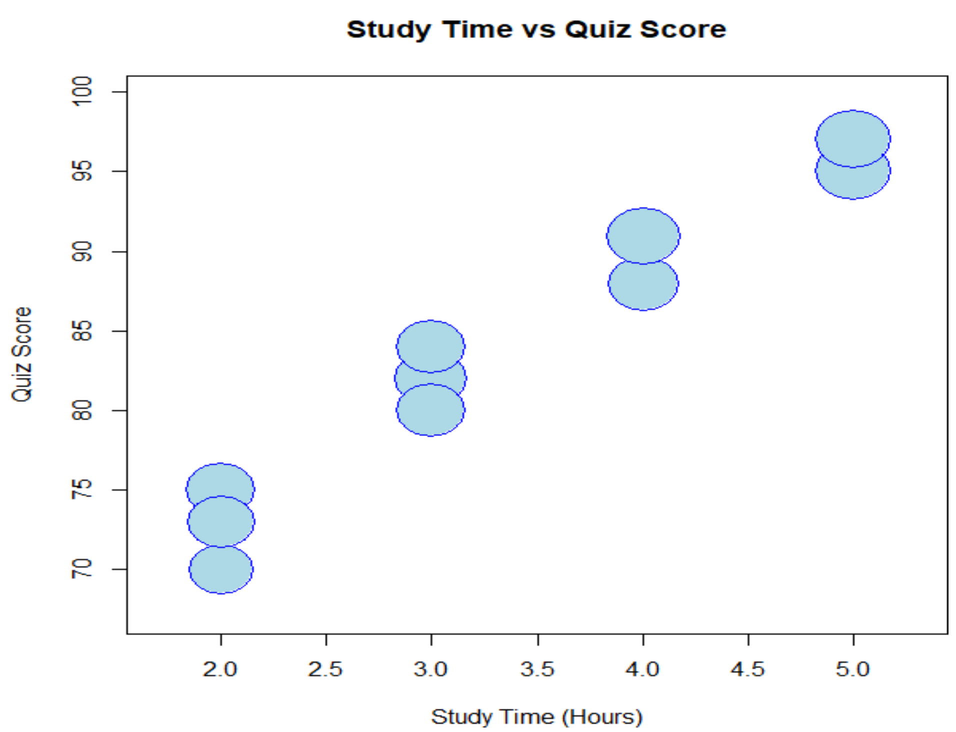


2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

Code:

```
# Study Time
Study_Time <- c(2,3,4,2,5,3,4,2,5,3)
# Quiz Score
Quiz_Score <- c(75,82,88,70,95,80,91,73,97,84)
# Attendance
Attendance <- c(90,95,92,85,98,90,96,88,99,91)
symbols(
Study_Time,
Quiz_Score,
circles=Attendance/15,
inches=0.25,
bg="lightblue",
fg="blue",
xlab="Study Time (Hours)",
ylab="Quiz Score",
main="Study Time vs Quiz Score"
)
```

Output:



3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

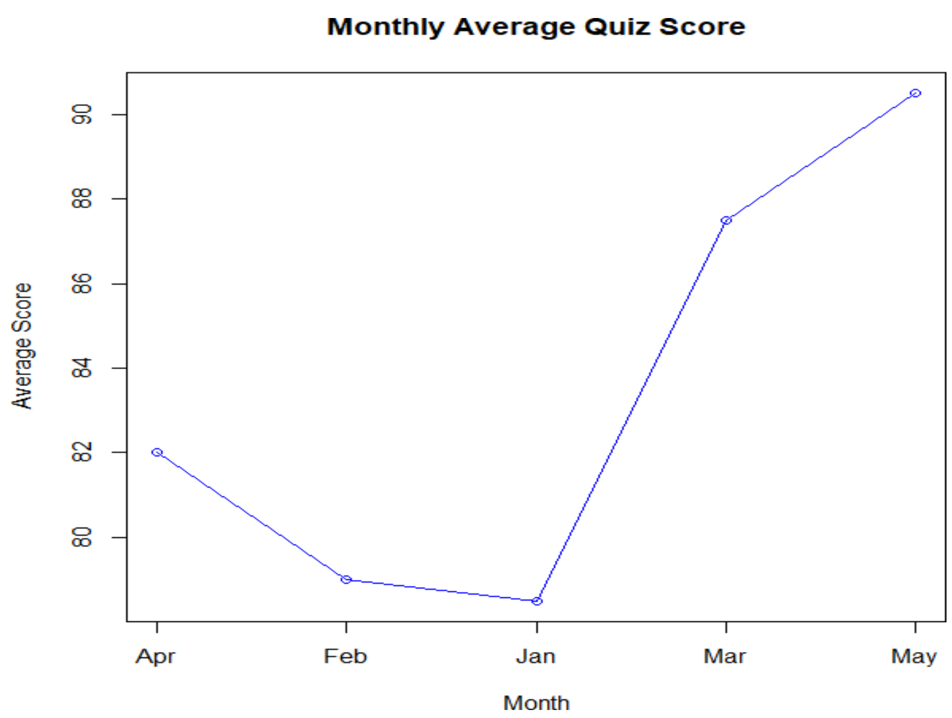
Code:

```
# Dataset

Month <- c("Jan","Jan","Feb","Feb","Mar","Mar","Apr","Apr","May","May")

Quiz_Score <- c(75,82,88,70,95,80,91,73,97,84)
monthly_avg <- aggregate(Quiz_Score ~ Month, FUN=mean)
print(monthly_avg)
plot(
monthly_avg$Quiz_Score,
type="o",
col="blue",
xaxt="n",
main="Monthly Average Quiz Score",
xlab="Month",
ylab="Average Score"
)
axis(1,
at=1:nrow(monthly_avg),
labels=monthly_avg$Month)
```

output:



Set 10 Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

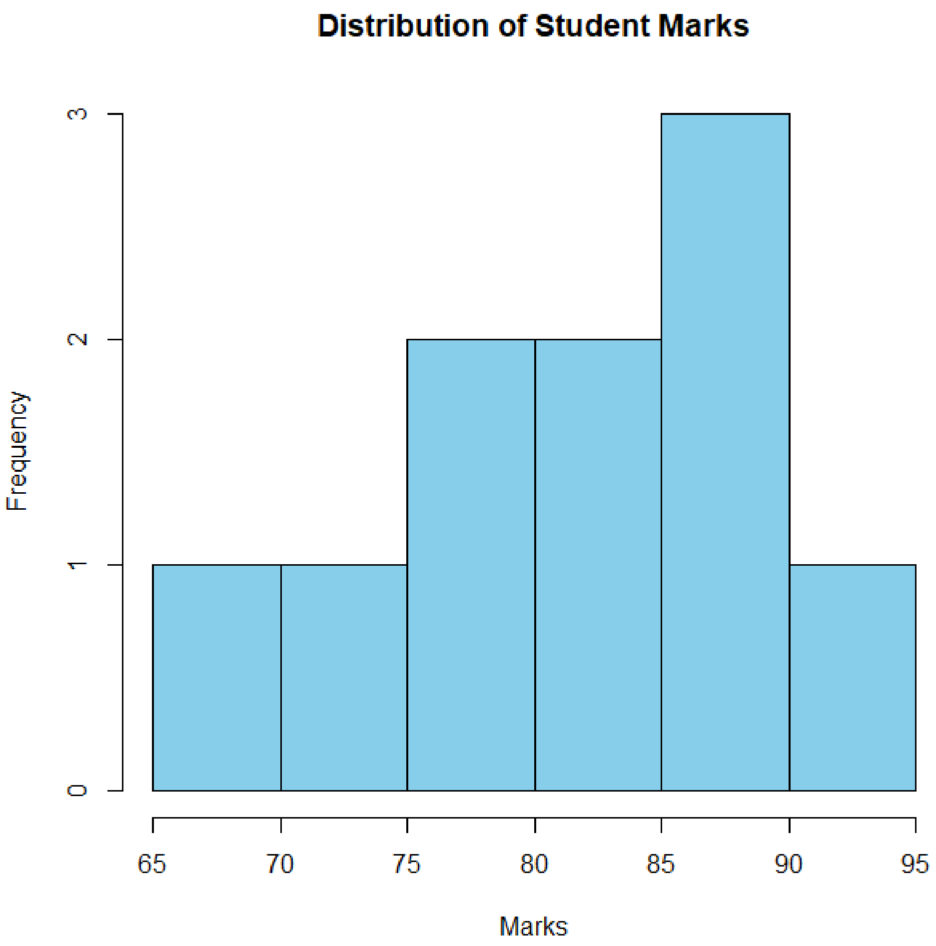
Code:

```
# Student Marks

Marks <- c(78,85,90,67,88,72,95,81,76,89)

hist(
Marks,
col="skyblue",
main="Distribution of Student Marks",
xlab="Marks",
ylab="Frequency"
)
```

Output:



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

Code:

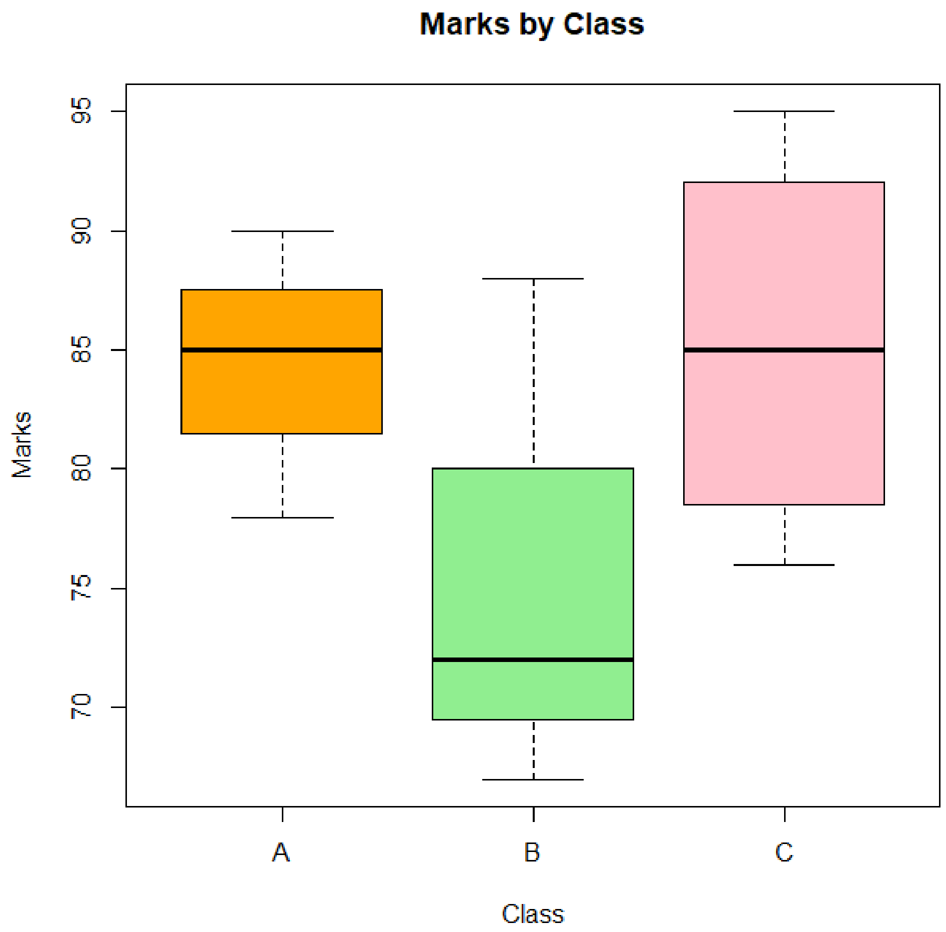
```
# Student Data

Class <- c("A","A","A","B","B","B","C","C","C","C")

Marks <- c(78,85,90,67,88,72,95,81,76,89)

boxplot(
Marks ~ Class,
col=c("orange","lightgreen","pink"),
main="Marks by Class",
xlab="Class",
ylab="Marks"
)
```

Output:



3. Build a table to show the survey response data for each survey. Label the table elements.

Code:

```
# Study Hours and Marks

Study_Hours <- c(2,3,5,1,4,2,6,3,2,5)

Marks <- c(78,85,90,67,88,72,95,81,76,89)

plot(
Study_Hours,
Marks,
pch=19,
col="red",
main="Study Hours vs Marks",
xlab="Study Hours",
ylab="Marks"
)

abline(lm(Marks ~ Study_Hours),col="blue",lwd=2)
```

output: